

Harrow Bay Offshore Wind – Wind Resource Assessment

Revision 2 – basis for financial close

Sofie Lindqvist · 21 November 2025 · HB-MET-RPT-0003

Executive summary

This report presents the long-term wind resource at the Harrow Bay Offshore Wind site, 38 km off the coast in the UK North Sea, based on a 24-month floating LiDAR campaign (LiDAR buoy HB-FLS-01, March 2021 to February 2023) correlated with 22 years of ERA5 reanalysis data. The long-term mean wind speed at the hub height of the Halden HTS-12 turbine (140 m) is **10.1 m/s**, with a Weibull scale parameter $A = 11.4$ m/s and shape $k = 2.18$. The prevailing wind direction is south-west to west.

This revision (Rev 2) supersedes the pre-consent assessment and supports the final energy yield assessment used for financial close on 12 February 2026. The Lenders' Technical Adviser reviewed the report without material findings.

Parameter	Value
Measurement period	2021-03-01 to 2023-02-28 (24 months)
Measurement device	Floating LiDAR HB-FLS-01, Stage 3 (OWA roadmap) validated
Data availability (140 m)	93.8 %
Measured mean wind speed at 140 m	10.3 m/s
Long-term reference	ERA5, 2001–2022, node 54.5° N 1.0° E
Long-term mean wind speed at 140 m	10.1 m/s
Wind shear exponent (100–140 m)	0.085
Mean turbulence intensity at 15 m/s	6.1 %
Air density (hub height)	1.226 kg/m³

Table 1 – Key wind resource results



Floating LiDAR buoy HB-FLS-01 during the 2021–2023 campaign

1. Measurement campaign

The floating LiDAR was deployed at the centre of the eastern array area in 31 m water depth. Service visits took place every six months. Two outages (October 2021, 11 days, power system fault; January 2023, 6 days, storm damage to the solar panels) are the main reasons for the reduced availability. Data were filtered for buoy motion, low carrier-to-noise ratio and icing of the optics; no gap filling was applied before the long-term correction.

2. Long-term correction

Measure-correlate-predict (MCP) was performed with a variance ratio method on hourly data, by 12 direction sectors. ERA5 showed the highest correlation ($R^2 = 0.93$) of the four reference datasets tested; MERRA-2 ($R^2 = 0.89$) was used as a cross-check. The measurement period was 1.6 % windier than the long-term average, which is why the long-term mean is lower than the measured value.

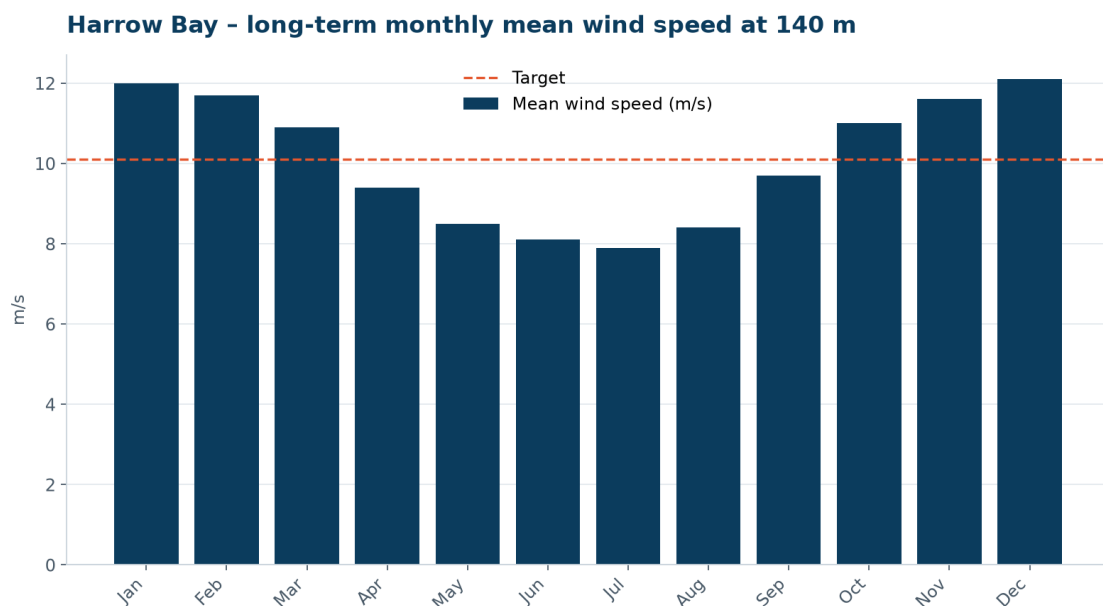


Figure 1 – Long-term monthly mean wind speed at 140 m

The seasonal pattern is pronounced: winter months (December to February) are around 50 % windier than summer months. This matters for construction planning – turbine installation from MV Sea Atlas is planned for the calmer window from March 2027 – and for the PPA with Arcturus Steel (WE-PPA-2026-004), which is pay-as-produced and therefore exposed to seasonal profile.

3. Wind speed distribution

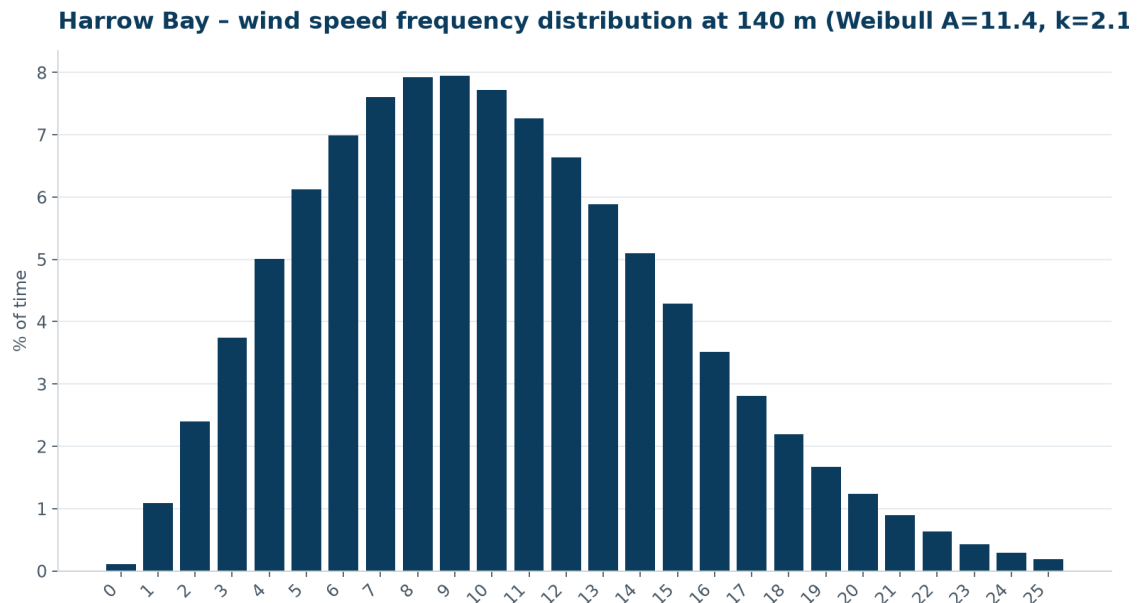


Figure 2 – Long-term frequency distribution at hub height

The fitted Weibull distribution matches the binned data well (RMSE 0.11 %). Wind speeds above the HTS-12 cut-out of 30 m/s occur less than 0.1 % of the time; high-wind ride-through is included in the power curve.

4. Wind direction

Sector	Frequency	Mean wind speed (m/s)
N (345–15°)	6.2 %	9.1
NNE–ENE	8.9 %	9.0
E–ESE	7.4 %	8.6
SE–S	13.8 %	9.8
SSW–WSW	31.6 %	11.0
W–WNW	22.1 %	10.6
NW–NNW	10.0 %	9.4

Table 2 – Directional distribution (7 sectors for readability; analysis in 12 sectors)

5. Extreme winds and turbulence

The 50-year 10-minute mean wind speed at hub height is 44.6 m/s (Gumbel method on ERA5 annual maxima, calibrated to measurements), within the IEC Class I envelope of the HTS-12. Ambient turbulence is low (6.1 % at 15 m/s); the effective turbulence including wakes is assessed in the load calculations by Halden Turbine Systems.

6. Uncertainty

Component	Uncertainty in wind speed
Measurement (floating LiDAR, Stage 3)	2.3 %
Long-term correlation	1.5 %
Long-term period representativeness	1.2 %
Vertical extrapolation (none needed at 140 m)	0.5 %
Horizontal extrapolation across the site	1.6 %
Total	3.4 %

7. Conclusions

- Long-term hub-height mean wind speed 10.1 m/s, confirmed as the basis for the P50/P90 yield assessment.
- No need for additional measurements before COD; the operational SCADA data will be used for the first reconciliation in 2029.
- Recommendation: retain the LiDAR buoy for a 12-month post-construction campaign to measure the cluster wake from neighbouring projects.

Prepared by Sofie Lindqvist (Head of Engineering). Checked by Oliver Grant. Reviewed by the Lenders' Technical Adviser.